

Generation Interconnection System Impact Study Report

For Queue Project Q18-0977

Blue Ravine-School 230 kV

July 28, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0977. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Blue Ravine-School 230 kV substation in the Indiana area. The request was received on April 08, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 381 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$148,820,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Blue Ravine-School-Henry County 230.0 kV line	Blue Ravine-School-Desoto 230.0 kV	195.6	507	48.8
Henry County 500/230 kV Autotransformer	East Methuen-Lonetree 500.0 kV	184.7	894	33.5
Sabanera-Murdock 230.0 kV	Henry County-East Methuen 500.0 kV Line	151.0	731	36.9
Desoto-Blue Ravine-School 230.0 kV	Henry County Substation-Blue Ravine-School 230.0 kV	140.9	643	41.8
Gray'S Flat-Baraga Vlg 230.0 kV	Madeline-E Methuen 230.0 kV	128.1	1,020	13.5
15Th Avenue S-Lucent Substation 230.0 kV Line	Omni-Hackberry-Norwich 500.0 kV Line	126.4	616	6.1
Henry County-Murdock 230.0 kV	E Methuen-Henry County 500.0 kV Line	126.3	848	23.7
15Th Avenue S-Norwich 230.0 kV	Omni-Hackberry-Norwich 500.0 kV	125.9	630	6.1
GrayS Flat-East Methuen 230.0 kV	East Methuen-Madeline 230.0 kV line	125.7	1,052	13.5
Lonetree 500/230kV TX	E Methuen-Lonetree 500.0 kV	124.9	811	7.0
Madeline-Baraga Vlg 230.0 kV	E Methuen Substation-Gray'S Flat 230.0 kV	121.8	687	10.2
James De Young Substation-Carley Road 230.0 kV	E Methuen 500 kV/230 kV XFMR	116.7	820	10.0
Larson Street Substation-Corridor 230.0 kV	Beaver-Trego Substation 500.0 kV	115.1	851	6.7
Truman-Lookingglass Substation 230.0 kV line	Trego-Beaver 500.0 kV	114.0	911	6.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Winthrop-Hebo 230.0 kV	Hebo-Barrett 230.0 kV	113.5	761	8.2
Waldo-Truman 230.0 kV L	Trego-Beaver 500.0 kV	112.2	949	6.7
Muskogee-Montpelier Electric Generating Sta 500.0 kV	Sabanra-Lonetree 500.0 kV L	112.0	3,404	28.8
Adirondack-James De Young 230.0 kV	St. Lawrence-James De Young 230.0 kV	112.0	995	9.8
Joso-Eufaula 230.0 kV	Beaver-Eufaula 500.0 kV	109.6	783	6.2
East Methuen 500-230 kV Transformer Bank	Trego Substation-Beaver 500.0 kV	109.4	2,263	21.5
Norwich Substation-Larson Street 230.0 kV	Norwich-Barrett 500.0 kV	109.3	1,299	10.6
Treat-Mouton 230.0 kV	E Paso-Ravenna Substation 500.0 kV	109.1	575	5.2
Imperial-Gruver 230.0 kV line	Eufaula-Imperial 500.0 kV	109.1	574	5.2
Corridor-Lookingglass 230.0 kV line	Trego-Beaver 500.0 kV	108.9	926	6.7
Hebo-Burris Pit 230.0 kV	Trego-Beaver Substation 500.0 kV	108.2	1,191	10.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Henry County-Blue Ravine-School Substation 230.0 kV	Rebuild line with higher-capacity conductor	\$31,261,000	71% (\$22,218,000)
Henry County 500 kV/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$33,189,000	36% (\$12,080,000)
Sabanera-Murdock 230.0 kV	Rebuild line with higher-capacity conductor	\$36,160,000	48% (\$17,303,000)
Desoto-Blue Ravine-School 230.0 kV	Rebuild line with higher-capacity conductor	\$28,344,000	62% (\$17,718,000)
Baraga Vlg-Gray'S Flat 230.0 kV L	Rebuild line with higher-capacity conductor	\$17,817,000	13% (\$2,294,000)
15Th Avenue S-Lucent 230.0 kV	Rebuild line with higher-capacity conductor	\$14,578,000	14% (\$1,966,000)
Henry County-Murdock 230.0 kV	Rebuild line with higher-capacity conductor	\$14,450,000	74% (\$10,663,000)
Norwich-15Th Avenue S 230.0 kV Line	Rebuild line with higher-capacity conductor	\$15,492,000	14% (\$2,089,000)
GrayS Flat-East Methuen 230.0 kV	Rebuild line with higher-capacity conductor	\$49,093,000	13% (\$6,320,000)
Lonetree 500/230kV TX	Replace transformer with higher-rated unit	\$48,694,000	26% (\$12,483,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Madeline-Baraga Vlg 230.0 kV	Rebuild line with higher-capacity conductor	\$18,957,000	13% (\$2,516,000)
James De Young Substation-Carley Road 230.0 kV	Rebuild line with higher-capacity conductor	\$31,166,000	12% (\$3,687,000)
Corridor-Larson St Substation 230.0 kV Line	Rebuild line with higher-capacity conductor	\$12,435,000	10% (\$1,285,000)
Lookingglass-Truman 230.0 kV	Rebuild line with higher-capacity conductor	\$22,899,000	10% (\$2,366,000)
Winthrop-Hebo 230.0 kV	Rebuild line with higher-capacity conductor	\$14,228,000	15% (\$2,191,000)
Waldo-Truman 230.0 kV L	Rebuild line with higher-capacity conductor	\$20,520,000	10% (\$2,121,000)
Muskogee-Montpelier Electric Generating Sta 500.0 kV	Rebuild line with higher-capacity conductor	\$105,058,000	16% (\$16,860,000)
Adirondack-James De Young 230.0 kV	Rebuild line with higher-capacity conductor	\$33,602,000	14% (\$4,742,000)
Joso-Eufaula 230.0 kV	Reconductor line	\$3,248,000	11% (\$363,000)
East Methuen 500-230 kV Transformer Bank	Replace transformer with higher-rated unit	\$28,595,000	13% (\$3,646,000)
Norwich-Larson St 230.0 kV	Reconductor line	\$4,429,000	11% (\$482,000)
Mouton Substation-Treat 230.0 kV Line	Reconductor line	\$8,512,000	12% (\$1,030,000)
Imperial-Gruver 230.0 kV line	Reconductor line	\$5,178,000	11% (\$587,000)
Corridor-Lookingglass 230.0 kV line	Reconductor line	\$9,267,000	10% (\$958,000)
Hebo-Burris Pit 230.0 kV	Reconductor line	\$5,820,000	15% (\$852,000)
	Total allocated to project		\$148,820,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.